

Somatotype of National Level Male Wrestlers

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ABSTRACT

The purpose of the present study is to investigate the somatotype of male wrestlers. A total of 61 male national level wrestlers were examined at SAI NS NIS Patiala during 2017-18. The anthropometric parameters measured for this research were height, weight, diameters of humerus & Femur; girths of upper arm normal & flex and calf; skinfolds of biceps, triceps, subscapular, supraspinale and calf, which has been taken by using standard instruments and standard techniques (Ross et al, 1978). Mean value, standard deviation (SD) and t test were used for comparative purpose. The means value of decimal age, height, weight, and height-weight ratio of male wrestlers of combined category were 29.15 ± 3.90 years, 169.43 ± 7.47 cm, 76.20 ± 11.96 kg and 40.10 ± 1.18 , respectively. Somatotype of the male wrestlers combined category has been found as endomorphic mesomorph, with mean of 4.22 – 6.90 – 1.01.

Key words

Anthropometric parameters, height-weight ratio and somatotype

INTRODUCTION

The athletes anthropometric dimensions, reflecting body shape, proportionally, and composition (Carter, 1970, 1984), plays a significant role in determining the potential for success in sport (Barttinelli, 1990). The wrestlers are characterised by attributes of brute strength, by being massively muscled, short legged and stationary type of athletes. It is contended that the wrestlers are built like weightlifters and Football players, with shorts necks,

powerful shoulders, wide trunks, short legs and great muscular strength (Sodhi, 1991).

Heavier weight category shows tendencies towards endo-mesomorphy; whereas, the lighter weight category, to balanced mesomorphy (Carter & Heath, 1990). Jagiello and Kruszewski (2009) found that a characteristics trait of wrestlers, in heavier weight category was also big massiveness of elbow, knee and pelvis width and big diameters of forearm and shin. Moreover, showed an

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increased fat content in heavy weight categories (20%), middle categories (18%) and light categories (14.9%). The purpose of the present study was to investigate the somatotype of national male wrestlers and tries to find out the difference between Sodhi and Sidhu 1984 report and present study.

METHODOLOGY

The present cross-sectional study has been conducted on 61 national level male wrestlers, age range from 20 to 35 years. The data were collected at SAI NS NIS Patiala during 2017-18. They were divided into three category viz. light class i.e., 57, 61, 65 & 70 kg (N=27), medium class i.e., 74, 79 & 86 kg (N=27) and heavy class i.e., 92, 97 & 100+ kg (N=7). The selected anthropometric parameters for this research were height, weight, diameters of humerus & femur; girths of upper arm normal & flex and calf; skinfolds of biceps, triceps, subscapular, supraspinale and calf, which has been taken by using standard instruments and Standard techniques (Ross et al, 1978). Mean value, standard deviation (SD) and t-test for comparison has been conducted. The formula to calculate height weight ratio (HWR) is mentioned as below:

$$\text{HWR} = \text{Height (cm)} / \sqrt[3]{\text{Weight (kg)}}$$

Somatotype components were computed according to computer equations of Carter (1980).

$$\text{Endomorphy} = -0.7182 + 0.1451(X) - 0.00068(X^2) + 0.0000014(X^3)$$

Where, X= sum of triceps, subscapular and supraspinale skinfolds.

For obtaining height corrected endomorphy, X is multiplied with 170.18 and divided by height of subject.

$$\text{Mesomorphy} = (0.858 \times \text{Humerus breadth}) + (0.601 \times \text{femur breadth}) + (0.188 \times \text{skinfold-corrected arm girth}) + (0.161 \times \text{skinfold-corrected calf girth}) - (0.131 \times \text{height}) + 4.5$$

Where, skinfold-corrected arm girth=flexed arm girth-triceps skinfold (cm) corrected calf girth = Max. calf girth-medial calf skinfold (cm)

$$\text{Ectomorphy} = \text{HWR} \times 0.732 - 28.58, \text{ where HWR} = \text{height/cube root of weight}$$

If $\text{HWR} < 40.75$ but > 38.25 , then $\text{ectomorphy} = \text{HWR} \times 0.463 - 17.63$ If $\text{HWR} \leq 38.25$, then $\text{ectomorphy} = 0.1$

RESULTS & DISCUSSION

The descriptive statistics, i.e., mean value, standard deviation and ranges of anthropometric parameters of male wrestlers combined category has been shown in Table 1. The decimal age in year's ranges from 20.35 to 35.92, with mean value 29.15 ± 3.90 . The mean value of height is 169.43 ± 7.47 , which ranges from 147 to 186.30. The mean of weight is 76.20 ± 11.96 , which ranges from 56.10 to 108.60 and mean of height-weight ratio is 40.10 ± 1.18 . Somatotype of the male wrestlers combined category has mean of endomorphy 4.22 ± 1.17 , mesomorphy 6.90 ± 0.96 and ectomorphy 1.01 ± 0.57 which falls under the somatotype classification of endomorphic mesomorph. Kroll's wrestlers had a

mean height of 174.8 cm, a mean weight of 73.1 kg and mean somatotype of his subjects was 2.7 – 5.0 – 3.8 (Kroll, 1954). The wrestler of Mexico Olympics had an average somatotype of 2.4 – 6.4 – 1.6 (de Garay et al, 1974).

Table-1: Descriptive Statistics of Male Wrestlers Combined Category (N=61)

Anthropometric Parameters	Min	Max	Mean	SD
Decimal Age (Years)	20.35	35.92	29.15	3.90
Height (cm)	147.00	186.30	169.43	7.47
Weight (kg)	56.10	108.60	76.20	11.96
Height-weight ratio	35.71	43.34	40.10	1.18
Endomorphy	1.87	7.24	4.22	1.17
Mesomorphy	4.82	10.90	6.90	0.96
Ectomorphy	0.10	3.15	1.01	0.57

Table-2: Descriptive Statistics of Male Wrestlers Category wise (N=61)

Anthropometric Parameters	Light Class (N=27)		Medium Class (N=27)		Heavy Class (N=7)	
	Range	Mean±SD	Range	Mean±SD	Range	Mean±SD
Decimal Age (years)	20.35-35.00	28.33±3.65	22.20-35.61	29.23±4.06	25.93-35.92	31.97±3.25
Height (cm)	147-170.70	163.64±5.11	162.00-179.80	172.74±4.94	168.20-186.30	179.03±5.66
Weight (kg)	56.10-71.80	65.56±4.53	73.00-89.50	81.16±5.32	91.60-108.60	98.09±6.21
Height-Weight ratio	38.40-43.34	40.61±0.95	37.62-41.99	39.92±1.04	35.71-40.29	38.85±1.52
Endomorphy	1.87-6.29	3.89±1.04	2.46-7.24	4.29±1.20	4.05-6.93	5.26±0.99
Mesomorphy	4.82-7.47	6.41±0.64	5.63-8.48	7.10±0.81	7.03-10.90	8.04±1.36
Ectomorphy	0.15-3.15	1.24±0.55	0.10-2.15	0.91±0.53	0.10-1.02	0.53±0.34

Table 2 shows the descriptive statistics of male wrestler's category wise i.e. light class, medium class and heavy class. It has been observed that the mean decimal age increased from light class to heavy class. Similarly the mean value of height, weight, somatotype 1st & 2nd component viz. endomorphy and mesomorphy also increase with the increase of weight

category. Thus, it can be noted that weight plays a major role in the weight category games like Wrestling. On the other hand anthropometric parameters like height-weight ratio and 3rd component of somatotype (ectomorphy) shows decrease in mean value with increase in weight category. This signifies the proportion and deposition of fat with respect to weight.

Table-3 a : Comparison of Anthropometric Parameters of Present Study with Sodhi and Sidhu, 1984 (Light Class)

Anthropometric Parameters	Present study (N=27)	Sodhi & Sidhu, 1984 (N=12)	t	p value
	Mean $\pm$ SD	Mean $\pm$ SD		
Decimal Age (years)	28.33 $\pm$ 3.65	22.4 $\pm$ 3.8	4.62**	0.0001
Height (cm)	163.64 $\pm$ 5.11	159.5 $\pm$ 6.8	2.05*	0.0481
Weight (kg)	65.56 $\pm$ 4.53	55.8 $\pm$ 4.7	6.14**	0.0001
Endomorphy	3.89 $\pm$ 1.04	2.46 $\pm$ 0.62	4.40**	0.0001
Mesomorphy	6.41 $\pm$ 0.64	4.83 $\pm$ 0.81	6.55**	0.0001
Ectomorphy	1.24 $\pm$ 0.55	2.04 $\pm$ 0.75	3.74**	0.0006

*significant at 0.05 level; ** significant at 0.001 level

Table 3 a, shows comparison of anthropometric parameters of male wrestlers of present study and Sodhi and Sidhu, 1984, for light class. It has been found that there is significant difference in decimal age, weight, endomorphy,

mesomorphy and ectomorphy at 0.001 level ($p < 0.0001$), and height has been observed statistically significant difference at 0.05 level ($t = 2.05$, $p = 0.0481$).

Table-3 b : Comparison of Anthropometric Parameters of Present Study with Sodhi and Sidhu, 1984 (Medium Class)

Anthropometric Parameters	Present study (N=27)	Sodhi & Sidhu, 1984 (N=12)	t	p value
	Mean $\pm$ SD	Mean $\pm$ SD		
Decimal Age (years)	29.23 $\pm$ 4.06	22.9 $\pm$ 3.1	5.37**	0.0001
Height (cm)	172.74 $\pm$ 4.94	170.7 $\pm$ 6.7	1.14	0.2589
Weight (kg)	81.16 $\pm$ 5.32	72.8 $\pm$ 7.2	4.36**	0.0001
Endomorphy	4.29 $\pm$ 1.20	3.62 $\pm$ 1.09	1.82	0.0746
Mesomorphy	7.10 $\pm$ 0.81	5.34 $\pm$ 0.89	6.63**	0.0001
Ectomorphy	0.91 $\pm$ 0.53	1.47 $\pm$ 0.76	2.84*	0.0069

*significant at 0.05 level; ** significant at 0.001 level

Table 3 b, shows the comparison of anthropometric parameters of male wrestlers of present study and Sodhi and Sidhu, 1984, for medium class. Statistically significant difference has been observed in decimal age,

weight and mesomorphy at 0.001 level ($p < 0.0001$), and ectomorphy found to be statistically significant difference at 0.05 level ($t = 2.84$, $p = 0.0069$). There is no significant difference in height and endomorphy.

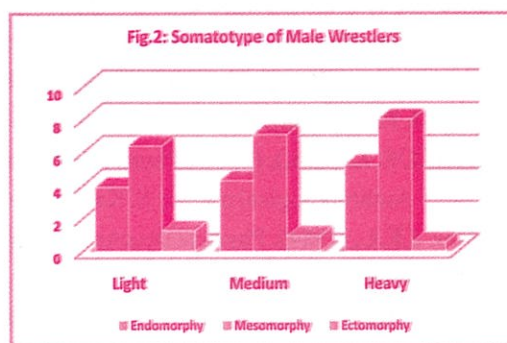
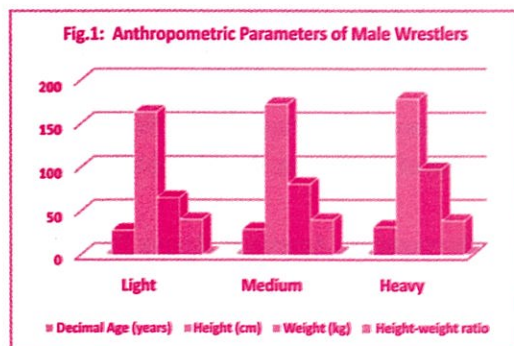
Table-3 c : Comparison of Anthropometric Parameters of Present Study with Sodhi and Sidhu, 1984 (Heavy Class)

Anthropometric Parameters	Present study (N=7)	Sodhi & Sidhu, 1984 (N=12)	t	p value
	Mean $\pm$ SD	Mean $\pm$ SD		
Decimal Age (years)	31.97 $\pm$ 3.25	28.1 $\pm$ 5.9	1.58	0.1334
Height (cm)	179.03 $\pm$ 5.66	179.9 $\pm$ 6.2	0.29	0.7682
Weight (kg)	98.09 $\pm$ 6.21	93.8 $\pm$ 9.2	1.08	0.2957
Endomorphy	5.26 $\pm$ 0.99	5.23 $\pm$ 1.68	0.04	0.9666
Mesomorphy	8.04 $\pm$ 1.36	6.18 $\pm$ 0.72	3.81**	0.0015
Ectomorphy	0.53 $\pm$ 0.34	1.00 $\pm$ 0.59	1.90	0.0752

*significant at 0.05 level; ** significant at 0.001 level

Comparison of anthropometric parameters of male wrestlers of present study and Sodhi and Sidhu, 1984, for heavy class has been shown in Table 3 c. Statistically significant difference has been observed in mesomorphy at 0.001 level ($t = 3.81$, $p = 0.0015$). Statistically no significant difference has been observed in other anthropometric parameters.

Fig. 1 and fig. 2, depicts the bar diagrammatic representation of anthropometric parameters and somatotype of male wrestlers with respect to weight category respectively.



CONCLUSION

It has been investigated from the present study that means value of decimal age, height, weight, and height-weight ratio of male wrestlers of combined category were 29.15 ± 3.90 , 169.43 ± 7.47 , 76.20 ± 11.96 and 40.10 ± 1.18 , respectively. Somatotype of the male wrestlers combined category has been found as endomorphic mesomorph with mean of $4.22 \pm 1.17 - 6.90 \pm 0.96 - 1.01 \pm 0.57$.

It is noted that all anthropometric parameter except height weight ratio and ectomorphy increases with the increase

of weight category from light to heavy class. The mean somatotype of light, medium and heavy class were 3.89 – 6.41 – 1.24, 4.29 – 7.10 – 0.91 and 5.26 – 8.04 – 0.54. Somatotype classification of all classes falls under endomorphic mesomorph, endomorphy; and mesomorphy shows higher with increase in weight category.

Further, by comparing

anthropometric parameters of present study with Sodhi and Sidhu, 1984 report, statistically significant differences, in all parameters, have been observed for light class male wrestlers. For medium class significant differences were found in decimal age, weight, mesomorphy and ectomorphy. Whereas for heavy class statistically significant difference has been observed only in mesomorphy.

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